PROGRAM:

AIES-Exp_3.py - C:/Users/ABBIRAMY/OneDrive/Documents/AIES_LAB/AIES-Exp_3.py (3.14.2)

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```
from itertools import permutations
words = ["SEND", "MORE"]
result = "MONEY"
letters = sorted(set("".join(words) + result))
def word_value(word, mapping):
    return int("".join(str(mapping[ch]) for ch in word))
def solve():
    for digits in permutations(range(10), len(letters)):
        mapping = dict(zip(letters, digits))
        # leading letters cannot be zero
        if any(mapping[w[0]] == 0 for w in words + [result]):
            continue
        if sum(word_value(w, mapping) for w in words) == word_value(result, mapping):
            return mapping
    return None
mapping = solve()
print("Letter -> Digit:", mapping)
print(f"  {word_value('SEND', mapping)}")
print(f"+ {word_value('MORE', mapping)}")
print("-------")
print(f"{word_value('MONEY', mapping)}")
```

OUTPUT:

IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/ABBIRAMY/OneDrive/Documents/AIES_LAB/AIES-Exp_3.py =====
Letter -> Digit: {'D': 7, 'E': 5, 'M': 1, 'N': 6, 'O': 0, 'R': 8, 'S': 9, 'Y': 2}
  9567
+ 1085
-------
10652
>>>
```